

Test 7 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.AR.3.1 easy	3:8	A part-to-whole ratio compares one part to the total of all the parts.
2	MA.6.NSO.1.3 moderate	5, -5	Both 5 and -5 are 5 units from 0.
3	MA.6.GR.1.3 moderate	(5, -3), (5, 2), (5, 6)	Every point on a vertical line has the same x-coordinate.
4	MA.6.AR.1.1 moderate	$5x + 2$, $2 + 5x$, $x \cdot 5 + 2$	Addition can be written in either order, but the coefficient and the constant cannot be combined.
5	MA.6.NSO.3.5 easy	$\frac{3}{5}$	A percent means hundredths, so write it over 100 and simplify.
6	MA.6.NSO.1.2 moderate	-11, 11	Below zero is negative, and the opposite of a negative number is positive.
7	MA.6.DP.1.4 easy	8	The range is the greatest value minus the least value shown on the plot.
8	MA.6.GR.2.2 moderate	15×5 , 5×15 , 75×1	The area of a parallelogram is the base times the perpendicular height.
9	MA.6.AR.2.2 moderate	$x + 9 = 46$, $x - 9 = 28$, $9 + x = 46$	Substitute 37 for x and keep the equations that stay balanced.
10	MA.6.AR.3.2 easy	60 miles	A unit rate tells how much for exactly one unit, so divide miles by hours.
11	MA.6.NSO.2.3 easy	\$-20.05	Find the cost of all the items first, then subtract that from the amount paid.
12	MA.6.AR.1.4 moderate	$8(x + 9)$, $8x + 72$, $72 + 8x$	Equivalent expressions have the same value for every value of x.
13	MA.6.GR.2.3 moderate	$8 \times 7 \times 2$, 56×2 , $2 \times 7 \times 8$	Volume is the product of the three edge lengths, which is also the area of the base times the height.
14	MA.6.GR.2.4 moderate	$2(5 \times 5) + 2(5 \times 6) + 2(5 \times 6)$, $2(5 \times 5 + 5 \times 6 + 5 \times 6)$, 170×1	Surface area adds the areas of all six faces, and opposite faces are equal.
15	MA.6.GR.1.1 moderate	(-3, -2), (-5, -4), (-1, -5)	A point is in Quadrant III when both coordinates are negative.
16	MA.6.NSO.4.2 easy	-4	Divide the absolute values, then decide the sign from the two signs.
17	MA.6.DP.1.3 easy	5	The interquartile range is the upper quartile minus the lower quartile.
18	MA.6.AR.3.3 easy	24	The total is the sum of the two parts in the same row.

#	Benchmark	Answer	Explanation
19	MA.6.AR.1.2 moderate	53, 58, 60	Substitute each value and check whether the inequality is true.
20	MA.6.DP.1.5 easy	4	Count how many values fall inside that interval, including both endpoints.
21	MA.6.DP.1.6 easy	28	The new value becomes the greatest value, so the range grows.
22	MA.6.AR.3.4 easy	200	Divide the part by the percent written as a decimal to find the whole.
23	MA.6.AR.2.4 easy	$\frac{1}{2}$	Divide the product by the known factor.
24	MA.6.NSO.4.1 easy	11	Subtracting a number is the same as adding its opposite.
25	MA.6.NSO.2.3 moderate	$-2\frac{3}{4}$	Multiply to find the ribbon used, then subtract from the whole roll.
26	MA.6.NSO.2.1 moderate	16.2	Multiply the unit price by the number of items.
27	MA.6.NSO.3.4 easy	50	Multiply all of the prime factors together.
28	MA.6.NSO.2.2 moderate	$13\frac{1}{2}$	Multiply the amount for one batch by the number of batches.
29	MA.6.AR.2.1 moderate	3, 5, 6	The symbol $\leq$ includes 6 itself.
30	MA.6.AR.3.5 easy	0.5	Divide the price by the number of bottles to compare unit prices.
31	MA.6.AR.1.3 moderate	$2x + 3$, $3 + 2x$, $2(x) + 3$	Substitute x and evaluate each expression using the order of operations.
32	MA.6.NSO.1.1 moderate	-5.5, $-1\frac{3}{4}$, $\frac{1}{4}$, 3	Order the numbers by their positions on a number line, from left to right.
33	MA.6.GR.2.1 moderate	$(14 \times 9) \div 2$, $0.5 \times 14 \times 9$, $14 \times 9 \times 0.5$	The area of a triangle is half the product of the base and the height.
34	MA.6.NSO.3.3 easy	$\frac{9}{16}$	Square the numerator and square the denominator.
35	MA.6.DP.1.2 easy	3	The mode is the value that appears most often.
36	MA.6.NSO.3.2 easy	36	Add inside the parentheses first, then multiply by the factor outside.
37	MA.6.NSO.3.1 easy	45	The next time they leave together is the least common multiple of the two intervals.
38	MA.6.NSO.1.4 moderate	$<$	Find both absolute values, then compare those distances.